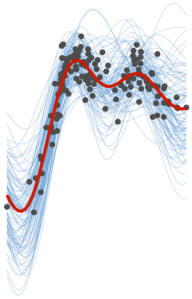
Haziq Jamil

haziq.jamil@kaust.edu.sa
https://haziqj.ml



View this CV online at *haziqj.ml/cv*

I am a Research Specialist working at King Abdullah University of Science and Technology (KAUST), currently on leave from my position as an Assistant Professor in Statistics at Universiti Brunei Darussalam (UBD). My expertise lie in statistical methodology and computation, with applications towards the natural and social sciences.

I am passionate about bringing data and technology to the forefront of our daily interaction with the world. Using the R programming language, I build statistical models for data analysis in order to gain insights, make predictions, and aid decision-making. I am frequently engaged in cross-disciplinary research collaborations and statistical consultations.

Research Interests

Latent variable models	Bayesian estimation
Composite likelihood methods	Integrated Nested Laplace Approximation (INLA)
Spatio-temporal models	Hedonic pricing models

Education

2014 — 2018 **PhD in Statistics**, London School of Economics and Political Science
Supervisors: Wicher Bergsma[1] and Irini Moustaki[2]
Thesis: *Regression modelling using priors depending on Fisher information covariance kernels (I-priors)*[3]

2013 — 2014 **MSc in Statistics (distinction)**, London School of Economics and Political Science

2006 — 2010 **BSc & Master in Mathematics, Operational Research, Statistics and Economics (first class honours)**, University of Warwick
Supervisor: David Firth[4]
Dissertation: *The analysis of paired comparison data using Bradley-Terry models with applications to football data*

Awards & honours

2023 **Teaching Excellence Award in the Sciences**, *Universiti Brunei Darussalam*

2020 **Arnold Zellner Thesis Award in Econometrics and Statistics (honourable mention)**[5], American Statistical Association (Business and Economic Statistics Section)

2013 — 2018 **In-Service Training Scheme Scholarship**, Public Service Commission, Prime Minister's Office, Brunei

2006 — 2010 **Supreme Commander of The Royal Brunei Armed Forces Scholarship**[6], Ministry of Defence, Brunei

Employment

2025 — present
Research Specialist
King Abdullah University of Science and Technology[7]
I conduct research on efficienct Bayesian computational methods for latent variable modeling within the BAYESCOMP group led by Prof. Håvard Rue.

2024 — 2027
Visiting Fellow
Department of Statistics, London School of Economics and Political Science[8]
The School's Visiting Appointments Scheme is a device to help Departments, Institutes and Centres invite staff who are either practitioners or academics in other institutions to conduct research and other activities.

2019 — present
Assistant Professor in Statistics
Mathematical Sciences, Faculty of Science, *Universiti Brunei Darussalam*[9]
[Leave of absence from August 2025]

I teach SM-1402 Basic Statistics, SM-2302 Software for Mathematicians, SM-4331 Advanced Statistics, SM-4335 Advanced Probability, and CU-0304 Sciences for University Study. I also supervise mathematics students' final year projects. In addition to my teaching duties, I am the current Postgraduate Coordinator (Mathematics). I previously held the post of Discovery Year[10] advisor (2019-2021).

2010 — 2014
2018 — 2019
Graduate Teaching Assistant
Department of Statistics, London School of Economics and Political Science[11]
I held various teaching roles ranging from secondary teaching to marking for masters (ST416 Multilevel Modelling, ST421 Development in Statistical Methods) and undergraduate level courses (ST312 Applied Statistics Project) at the LSE. In addition, I also held assistant invigilator and examiner roles for both the LSE and the University of London International Examinations respectively.

2010 — 2019
Research Officer
Centre for Science and Technology, Research and Development (CSTRAD), Ministry of Defence, Brunei[12]
I provided analytical insight into matters arising from strategic procurement activities of the Royal Brunei Armed Forces (RBAF). This entailed market research, cost-benefit analyses, and writing technical reports. I was a core team member for the procurement and delivery of 12 Sikorsky S-70i helicopters[13] (US$325M), and also the focal point for the development of tender evaluation criteria using the Analytic Hierarchy Process.

Service

2024 — 2026
Chair
Brunei R User Group[14]

2020 — present
Ad-hoc Reviewer
Annals of Applied Statistics[15]; Journal of the Royal Statistical Society: Series C[16]

2020 — present
Review Editor
Frontiers in Applied Mathematics and Statistics[17]

2020
Guest Lecturer
Universiti Teknologi Brunei[18]

Consultancy

2021 — 2022 **Statistical Modelling**
Ministry of Defence, Brunei
Project: Human performance optimisation

2017 — 2018 **Data Analyst**
Universiti Brunei Darussalam
Project: The unmet needs of cancer carers in Brunei Darussalam.

Grants and funding

2026 — 2029 **Advancing scalable Bayesian methods for latent variable modelling** (USD 900,000), KAUST Competetive Research Grant[19].
Role: Co-Investigator

2024 **R User Groups (RUGS) Program Grants** (USD 500), R Consortium[20]

2023 — 2024 **Predictive potential of graphical descriptors for QSPR/QSAR model learning of compounds** (~BND 15,700), Faculty of Science Research Grant, *Universiti Brunei Darussalam*.
Role: *Co-Principal Investigator*

2021 — 2024 **Extension of I-prior methodology to nominal response models** (~BND 7,000), Faculty of Science Research Grant, *Universiti Brunei Darussalam*.
Role: *Principal Investigator*

2021 — 2024 **Mathematical Modeling of Genetic Assimilation** (~BND 8,000), Faculty of Science Research Grant, *Universiti Brunei Darussalam*.
Role: *Co-Principal Investigator*

2021 — 2023 **Examining spillover effects in regional trading blocs** (~BND 17,000), School of Business and Economics Research Grant, *Universiti Brunei Darussalam*.
Role: *Member*

Research output

Articles (peer-reviewed)

2026 A19 **Bias-Reduced Estimation of Structural Equation Models**
H. Jamil, Y. Rosseel, O. Kemp, I. Kosmidis
Structural Equation Modeling: A Multidisciplinary Journal, 33(3), 376–392. https://doi.org/10.1080/10705511.2025.2610462

2025 A18 **Conditionally Autoregressive Models for House Price Data: Insights from a Comparative Simulation Study**
I. Puteri Kinasih, *H. Jamil*, E. Leong
Journal of Statistical Theory and Applications. https://doi.org/10.1007/s44199-025-00122-1

A17 **Leveraging sparse Gaussian processes for property price modelling and sustainable urban planning**
H. Jamil, F. Usop, H. M. Ramli
In: S. A. Abdul Karim & A. Baharum (Eds.), Intelligent Systems of Computing and Informatics in Sustainable Urban Development. Taylor and Francis/CRC Press. https://doi.org/10.1201/9781003518396-2

A16 **A spatio-temporal analysis of house prices in Brunei Darussalam**
H. Jamil
Quality & Quantity. https://doi.org/10.1007/s11135-025-02164-0

A15 **From archives to AI: Residential property data across three decades in Brunei Darussalam**
H. Jamil, A. Barizah Noorosmawie, H. Waezz Rabu, L. Abdul Razak
Data in Brief, 60, 111505. https://doi.org/10.1016/j.dib.2025.111505

2024 A14 **An Exploratory Investigation on Foreign Language Anxiety among English for Islamic Studies Learners in Brunei Darussalam**
N. Noorashid, *H. Jamil*
Electronic Journal of Foreign Language Teaching, 21(2) 211-234. https://doi.org/10.56040/nnhj2123

A13 **Pairwise likelihood estimation and limited-information goodness-of-fit test statistics for binary factor analysis models under complex survey sampling**
H. Jamil, I. Moustaki, C. Skinner
British Journal of Mathematical and Statistical Psychology, 78(1), 258–285. https://doi.org/10.1111/bmsp.12358

A12 **An Optimization Problem for Computing Predictive Potential of General Sum/Product-Connectivity Topological Indices of Physicochemical Properties of Benzenoid Hydrocarbons**
S. Hayat, A. Arfan, A. Khan, *H. Jamil*, M. J. F. Alenazi.
Axioms 13(6). https://doi.org/10.3390/axioms13060342

2023 A11 **Importance of Baseline Assessments: Monitoring of Sungai Brunei River's Water Quality**
O. Onifade, N. Shamsuddin, *H. Jamil*, D. T. C. Lai, S. Gödeke
H2Open Journal, 6(4) 518–534. https://doi.org/10.2166/h2oj.2023.168

A10 **Extremal hyper-Zagreb index of trees of given segments with applications to regression modeling in QSPR studies**
S. Hayat, M. Adil Khan, A. Khan, *H. Jamil*, M. Yasir Hayat Malik
Alexandria Engineering Journal, 80(1) pp.259–268

A09 **Statistical significance of valency-based topological descriptors for correlating thermodynamic properties of benzenoid hydrocarbons with applications**
S. Hayat, N. Suhaili, *H. Jamil*
Computational and Theoretical Chemistry, 114259. https://doi.org/10.1016/j.comptc.2023.114259

A08 **The domination number of the king's graph**
M. Arshad, S. Hayat, *H. Jamil*
Computational and Applied Mathematics, 42(251). https://doi.org/10.1007/s40314-023-02386-8

2022 A07 **Factors Affecting Readmission to a Drug Rehabilitation Centre**
F. Jali, E. Leong, *H. Jamil*
Journal of Population and Social Studies, 30, 591-601.

A06 **Iron and Manganese Mobilisation due to Dam Height Increase for a Tropical Reservoir in South East Asia**
S. Gödeke, *H. Jamil*, M. Schirmer, A. Bretzler, N. Shamsuddin, N. H. Mansor
Environmental Monitoring and Assessment, 194(5), 358.

A05 **Pairwise likelihood estimation for confirmatory factor analysis models with categorical variables and data that are missing at random**
M. Katsikatsou, I. Moustaki, *H. Jamil*
British Journal of Mathematical and Statistical Psychology, 75(2) 23-45.
🏆 Wiley Top Downloaded Article 2021

2021 A04 **Advocating Blended Learning for University Undergraduate Level Mathematical Instruction Beyond Covid-19**
H. Jamil, H. M. Ramli, E. Leong
In: Engineering and Sciences Teaching and Learning Activities: New Systems Throughout COVID-19 Pandemics, pp. 33-45. Ed. by Samsul Ariffin Abdul Karim and Saiful Azmi Hussain. Studies in Systems, Decision and Control, vol 381. Cham: Springer. ISBN: 978-3-030-79613-6.

A03 **Latent Class Analysis: Insights about design and analysis of schistosomiasis diagnostic studies**
A. Koukounari, *H. Jamil*, E. Erosheva, C. Shiff, I. Moustaki
PLOS Neglected Tropical Diseases, 15(2), e0009042.

2020 A02 **Bayesian Variable Selection for Linear Models Using I-Priors**
H. Jamil, W. Bergsma
In: Theoretical, Modelling and Numerical Simulations Toward Industry 4.0, pp. 107–132. Ed. by Samsul Ariffin Abdul Karim. Studies in Systems, Decision and Control, vol 319. Singapore: Springer. ISBN: 978-981-15-8986-7.

2018 A01 **Regression modelling using priors depending on Fisher information covariance kernels (I-priors)**
H. Jamil
PhD Thesis. London School of Economics and Political Science.
🎗 Arnold Zellner award, honourable mention (top 3)

Working papers (preprints and unpublished reports)

2026 W13 **Deterministic Leave-One-Cluster-Out Cross-Validation for Multilevel Bayesian Structural Equation Models**
M. Alhyari, *H. Jamil*, H. Montcho, H. Rue
arXiv:2609.00670 [stat.ME]

W12 **Informative Distance-Based Priors for Correlation Matrices Centred on a Target Reference**
A. Freni-Sterrantino, J. van Niekerk, E. Teixeira Krainski, D. Rustand, *H. Jamil*, H. Rue
arXiv:2607.18926 [stat.ME]

W11 **Implementation and Workflows for INLA-based Approximate Bayesian Structural Equation Modelling**
H. Jamil, H. Rue
arXiv:2604.00671 [stat.CO]

W10 **Approximate Bayesian Inference for Structural Equation Models Using Integrated Nested Laplace Approximations**
H. Jamil, H. Rue
arXiv:2603.25690 [stat.ME]

2025 W09 **A Bayesian factor analytic approach for handling missing data in spatial regressions models**
I. Puteri Kinasih, *H. Jamil*, E. Leong
10.13140/RG.2.2.30776.61448

2023 W08 **Additive interaction modelling using I-priors**
W. Bergsma, *H. Jamil*
arXiv:2007.15766 [math.ST]

W07 **Patterns of metabolic thermal independence in marine gastropods: A meta-analytical assessment**
M. Akram Ullah, D. Marshall, *H. Jamil*
Manuscript in preparation

2021 W06 **I think I understand: Investigating misconceptions regarding hypothesis test concepts among university students**
A. Zaini, *H. Jamil*, E. Leong
EdArXiv. https://doi.org/10.35542/osf.io/dj6x9

2020 W05 **Regression modelling with I-priors: With applications to functional, multilevel and longitudinal data**
W. Bergsma, *H. Jamil*
arXiv:2007.15766 [math, stat].

W04 **Investigating the effect of load carriage on soldiers' performances using Bayesian structural equation models**
H. Jamil, A. Salleh, L. Thieng Chan

2019 W03 **iprior: An R Package for Regression Modelling using I-priors**
H. Jamil, W. Bergsma
arXiv:1912.01376 [stat].

2017 W02 **Discussion contribution**
W. Bergsma, *H. Jamil*
TI Cannings, RJ Samworth (2017). Random-projection ensemble classification. JRSSB, 79.4, pp. 959-1035.

2010 W01 **Analysis of paired comparison data using Bradley-Terry models with applications to football data**
H. Jamil
Masters Thesis. University of Warwick.

Software

2026 S5 **R/INLAvaan: Approximate Bayesian Latent Variable Analysis**
H. Jamil


2024 S4 **R/bruneimap: Maps and Spatial Data of Brunei**
H. Jamil


2023 S3 **R/lavaan: Latent Variable Analysis**
Y. Rosseel, T. D. Jorgensen, L. De Wilde, D. Oberski, J. Byrnes, L. Vanbrabant, V. Savalei, E. Merkle, M. Hallquist, M. Rhemtulla, M. Katsikatsou, M. Barendse, N. Rockwood, F. Scharf, H. Du, *H. Jamil*, F. Classe
Role: Contributor. General maintenance, including but not limited to pairwise likelihood and bias-reduced estimation.


2021 S2 **R/iprior: Regression Modelling using I-priors**
H. Jamil


2020 S1 **stat-tables: Open-source statistical tables for use in examinations, tests, and assignments**
H. Jamil
release v0.2 downloads 1.2k

Talks

Further information regarding the following events are available at haziqj.ml/talk

2026 T37 **Centre for Educational Measurement (CEMO)**, University of Oslo, Norway
Approximate Bayesian inference for Structural Equation Models (SEM): The INLA approach

T36 **Graduate Statistics Seminar**, KAUST, Kingdom of Saudi Arabia
Approximate Bayesian inference for Structural Equation Models (SEM): The INLA approach

T35 **International Meeting of the Psychometric Society (IMPS)**, Yonsei University, Seoul, South Korea
Approximate Bayesian inference for Structural Equation Models (SEM): The INLA approach

T34 **Guest lecture for Calculus I (MAT183)**, Universiti Teknologi Mara (UiTM) Sarawak Branch
Calculus: Differentiation and its Applications (A statistical perspective)

T33 **School of Public Health Seminar**, Fudan University, Shanghai, China
Approximate Bayesian inference for Structural Equation Models: The INLA approach

T32 **International Guest Lecture (online)**, Universitas Islam Negeri (UIN) Mataram, Indonesia, Mathematics Education Programme
An Introduction to Item Response Theory (IRT) Models

T31 **Psychoco 2026**, Università di Padova, Italy
Approximate Bayesian inference for Structural Equation Models (SEM): The INLA approach

T30 **Social Statistics Seminar**, LSE, London, UK
Approximate Bayesian inference for Structural Equation Models (SEM): The INLA approach

2025 T29 **Graduate Statistics Seminar**, KAUST, Kingdom of Saudi Arabia
Bias-reduced estimation of structural equation models

T28 **Guest lecture for Calculus I (MAT183)**, Universiti Teknologi Mara (UiTM) Sarawak Branch
Calculus: Differentiation and its Applications (A statistical perspective)

T27 **Seminar talk on R programming**, Authority for Info-communications Technology Industry (AITI)
R for Data Science

2024 T26 **KAUST STAT Workshop 2024**, KAUST, Kingdom of Saudi Arabia
Leveraging INLA for estimating Bayesian latent variable models

T25 **Department of Statistics and Data Science Seminar**, NUS, Singapore
Weighted pairwise likelihood goodness-of-fit tests for binary factor models

	T24	**Mathematical Science Departmental Seminar**, UBD, Brunei Item Response Theory (IRT) models: Reducing bias in small samples
	T23	**International Meeting of the Psychometric Society (IMPS)**, Prague University of Economics and Business, Czech Republic Empirical bias-reducing adjustments for Item Response Theory (IRT) models
	T22	**Adolphe Quetelet Seminar Series**, Ghent University, Belgium Weighted pairwise likelihood and limited information goodness-of-fit tests for binary factor models
	T21	**Brunei R User Group Meetup**, UBD Startup Centre Analysing spatial data using R
2023	T20	**Social Statistics Seminar**, LSE, London, UK Spatio-temporal modelling of property prices in Brunei Darussalam
	T19	**Mathematical Science Departmental Seminar**, UBD, Brunei Pairwise likelihood goodness of fit tests for binary factor models
	T18	**International Meeting of the Psychometric Society (IMPS)**, University of Maryland, USA Pairwise likelihood goodness of fit tests for binary factor models
2022	T17	**Teaching and Learning Centre Workshop**, UBD, Brunei Quality Typography in Teaching Material: An interactive introduction to LaTeX
	T16	**Department of Statistics and Data Science Seminar**, NUS, Singapore Regression modelling using I-priors
	T15	**International Meeting of the Psychometric Society (IMPS)**, University of Bologna, Italy Selecting interaction effects in additive models using I-priors
2021	T14	**MINDEF Scholars Sharing Session**, Online Life After MINDEF
2020	T13	**IADA Seminar**, UBD, Brunei A latent variable model for maximal performance testing with dropouts for military applications
	T12	**Social Statistics Group Meetings**, LSE, London, UK Investigating the effect of load carriage on soldiers' performances using structural equation models
2019	T11	**UBDSBE Seminar Invitation**, UBD, Brunei Bayesian Variable Selection for Linear Models
	T10	**CSTRAD Sharing Session**, Ministry of Defence, Brunei Misconceptions in Demography
2018	T09	**FOS Research Seminar**, UBD, Brunei A Brief Guide to Variational Inference
	T08	**Inaugural Workshop on Data Science Theory and Practice**, LSE, London, UK Binary and Multinomial Regression using Fisher Information Covariance Kernels (I-priors)
	T07	**Social Statistics Group Meetings**, LSE, London, UK A Beginner's Guide to Variational Inference

2017 T06 **Turing Data Study Group**, The Alan Turing Institute, London, UK
Can data science help identify potential drivers of extremism?

T05 **International Meeting of the Psychometric Society (IMPS)**, University of Zürich, Switzerland
Regression Modelling with I-Priors

T04 **Department of Statistics PhD Presentation Event**, LSE, London, UK
Binary probit regression with I-priors

2016 T03 **Social Statistics Group Meetings**, LSE, London, UK
I-priors in Bayesian Variable Selection: From Reproducing Kernel Hilbert Spaces to Hamiltonian Monte Carlo

2015 T02 **Postgraduate Research Seminar**, LSE, London, UK
Two-stage Bayesian variable selection for linear models using I-priors

T01 **Department of Statistics PhD Presentation Event**, LSE, London, UK
Regression Modelling using I-priors

Skills

Programming, data analysis and visualisation using R, BUGS/JAGS, Stan, Stata, Mplus, and Mathematica

Reproducible research using Quarto, R Markdown, LaTeX, Git and AWS EC2

Web development using HTML, CSS, Hugo, GitHub Pages and Netlify

Made with the R package pagedown. The source code is available from github.com/haziqj/cv. Document last rendered on 2026-10-28 15:19:22.912639 [UTC].

Links

1. http://stats.lse.ac.uk/bergsma/↩
2. http://stats.lse.ac.uk/moustaki/↩
3. https://phd.haziqj.ml/↩
4. http://www2.warwick.ac.uk/fac/sci/statistics/staff/academic-research/firth/↩
5. https://community.amstat.org/businessandeconomicstatisticssection/new-item/new-item2↩
6. https://scholarship.mindef.gov.bn/Theme/Home.aspx↩
7. http://bayescomp.kaust.edu.sa/↩
8. http://lse.ac.uk/Statistics/↩
9. http://fos.ubd.edu.bn/↩
10. https://ubdmaths.github.io/dy/↩
11. http://lse.ac.uk/Statistics/↩

12. https://www2.mindef.gov.bn/cstrad/↩

13. https://news.lockheedmartin.com/2011-12-13-Sikorsky-Signs-with-Brunei-Ministry-of-Defence-for-Sale-of-S-70iTM-BLACK-HAWK-Helicopters↩

14. https://bruneir.github.io↩

15. https://imstat.org/journals-and-publications/annals-of-applied-statistics/↩

16. https://rss.onlinelibrary.wiley.com/journal/14679876↩

17. https://www.frontiersin.org/journals/applied-mathematics-and-statistics/↩

18. http://www.utb.edu.bn/↩

19. https://bayescomp.kaust.edu.sa/articles/2026/03/10/bayescomp-group-secures-crg-funding-revolutionize-latent-variable-modeling↩

20. https://www.r-consortium.org/all-projects/r-user-group-support-program↩